INF10025: TEST 2 REVIEW EXERCISES & SOLUTIONS

Comprehensive Practice Problems for Weeks 05, 06, and 07 | SQL, Relational Schema & ERD

SECTION 1: ERD & Relational Schema Conversion (Week 05)

Exercise 1.1: Converting a 1:M Relationship to Relational Schema

Scenario: Consider two entities: DEPARTMENT (DepartmentId, DeptName) and EMPLOYEE (EmpId, EmpName, Salary). A Department employs MANY Employees, but each Employee belongs to exactly ONE Department.

Task: Write the formal Relational Schema notation for both relations.

Solution & Rule:
• Foreign Key (FK) placement rule: In a 1:M relationship, the Foreign Key MUST be placed in the relation at the **MANY end** (the Child table).
• **DEPARTMENT** (DepartmentId, DeptName)
• **EMPLOYEE** (EmpId, EmpName, Salary, DepartmentId)
Note: DepartmentId in EMPLOYEE is a Foreign Key referencing DEPARTMENT(DepartmentId).

> ***SCHEMA NOTATION TRAP:*** *Always underline the Primary Key (PK) attribute. Foreign Key column names do NOT have to be identical to PK names, but MUST match data types.*

Exercise 1.2: SQL DDL Table Creation with Constraints

Task: Write an Oracle SQL CREATE TABLE statement for EMPLOYEE ensuring that EmpId is the Primary Key, Salary is mandatory, and DepartmentId references DEPARTMENT with a mandatory membership constraint.

```
CREATE TABLE EMPLOYEE (
   EmpId NUMBER PRIMARY KEY,
   EmpName VARCHAR2(50) NOT NULL,
   Salary NUMBER(10,2) NOT NULL,
   DepartmentId NUMBER NOT NULL,
   FOREIGN KEY (DepartmentId) REFERENCES DEPARTMENT(DepartmentId)
);
```

Exercise 1.3: DML Insert and NULL Value Handling

Question: Which of the following SQL statements correctly inserts a lecturer without an assigned Age?

A) INSERT INTO Lecturer VALUES (974, 'Robert Tipping', 'Null');
B) INSERT INTO Lecturer VALUES (974, 'Robert Tipping', Null);

Answer: B. In SQL, Null is a keyword representing an unassigned/unknown state. Writing 'Null' in quotes inserts a literal text string rather than a true SQL NULL value.

SECTION 2: Foreign Key Constraints & Multi-Table INNER JOINs (Week 06)

Exercise 2.1: Predicting Output Rows & Cartesian Product Prevention

Scenario: Table STUDENT has 100 rows. Table ENROLMENT has 500 rows. You run a query connecting both tables.

Case A: You execute SELECT * FROM STUDENT, ENROLMENT; (No JOIN condition). How many rows returned?

Case B: You execute SELECT * FROM STUDENT S INNER JOIN ENROLMENT E ON S.StudentId = E.StudentId; assuming all enrolments belong to existing students. How many rows returned?

Solution:
• **Case A Output:** 100 × 500 = **50,000 rows** (Cartesian Product / Cross Product).
• **Case B Output: 500 rows** (Matches each enrolment to its corresponding student).

Exercise 2.2: Writing Multi-Table JOIN Queries with Table Aliases

Task: Write a SQL query to display StudentName, SubjectCode, and Grade by joining STUDENT (S), ENROLMENT (E), and SUBJECT (Sub). Filter for subjects starting with 'INF'.

```
SELECT S.StudentName, Sub.SubjectCode, E.Grade
FROM STUDENT S
INNER JOIN ENROLMENT E ON S.StudentId = E.StudentId
INNER JOIN SUBJECT Sub ON E.SubjectCode = Sub.SubjectCode
WHERE Sub.SubjectCode LIKE 'INF%';
```

Exercise 2.3: Logical Operator Precedence (AND vs OR)

Problem: Evaluate the difference between these two WHERE clauses:
Query 1: WHERE Rating = 'PG' OR Rating = 'G' AND Runtime < 90
Query 2: WHERE (Rating = 'PG' OR Rating = 'G') AND Runtime < 90

Explanation: SQL evaluates AND before OR. Query 1 means 'All PG movies regardless of length, OR G movies under 90 mins'. Query 2 correctly restricts BOTH PG and G movies to under 90 minutes.

> ***REFERENTIAL INTEGRITY TRAP:*** *Deleting a record from a Parent table while child records exist causes ORA-02292 (child record found violation) unless CASCADE DELETE is specified.*

SECTION 3: Composite Keys, Aggregate Functions & GROUP BY / HAVING (Week 07)

Exercise 3.1: Composite Foreign Key Definition

Scenario: Table EMPLOYEE has a composite Primary Key (GivenName, Surname). Table PROJECT_ASSIGNMENT references this employee.

Task: Write the Foreign Key constraint in PROJECT_ASSIGNMENT.

```
FOREIGN KEY (EmpGivenName, EmpSurname)
   REFERENCES EMPLOYEE(GivenName, Surname)
```

Exercise 3.2: GROUP BY Syntax & Preventing Error ORA-00937

Problem: Why does the following query fail, and how do you fix it?
SELECT DepartmentId, JobTitle, AVG(Salary) FROM EMPLOYEE;

Diagnosis: Causes error ORA-00937: not a single-group group function because non-aggregated columns (DepartmentId, JobTitle) appear in SELECT without a GROUP BY clause.

Correct Query:

```
SELECT DepartmentId, JobTitle, ROUND(AVG(Salary), 2) AS AvgSal
FROM EMPLOYEE
GROUP BY DepartmentId, JobTitle;
```

Exercise 3.3: Differentiating WHERE vs HAVING Clauses

Task: Find departments where average salary exceeds $75,000, considering only full-time employees (Status = 'FT').

```
SELECT DepartmentId, ROUND(AVG(Salary), 2) AS AvgSalary
FROM EMPLOYEE
WHERE Status = 'FT' -- Step 1: Filter raw rows BEFORE grouping
GROUP BY DepartmentId -- Step 2: Group remaining rows
HAVING AVG(Salary) > 75000; -- Step 3: Filter aggregated groups
```

SECTION 4: Test 2 High-Yield Exam Practice Quiz (10 Questions)

Q1: 1. In a 1:M relationship between LECTURER (1) and SUBJECT (M), where is the Foreign Key placed?

A) In LECTURER
B) In SUBJECT
C) In a separate junction table
D) In both tables

Explanation: The FK always goes into the relation on the 'MANY' side.

Q2: 2. What happens if you run: SELECT ROUND(AVG(CreditPoints)) FROM SUBJECT; without a second argument in ROUND()?

A) Returns error ORA-00904
B) Rounds the average to the nearest whole integer (0 decimals)
C) Keeps 2 decimal places by default
D) Returns NULL

Explanation: Omitting the decimal parameter in ROUND() defaults to 0 decimal places.

Q3: 3. Which clause is used to filter aggregated group results after grouping has occurred?

A) WHERE
B) HAVING
C) ORDER BY
D) GROUP BY

Explanation: WHERE filters individual rows before grouping; HAVING filters groups after aggregate calculation.

Q4: 4. If a primary key in parent table 'BRANCH' is a Composite Key (StateCode, BranchNo), what must be true about the Foreign Key in table 'EMPLOYEE'?

A) It only needs to reference StateCode.
B) It must contain both StateCode and BranchNo as a composite FK.
C) Foreign Keys cannot reference composite primary keys.
D) It must be an AutoNumber field.

Explanation: A Foreign Key referencing a composite Primary Key must contain matching columns for all PK parts.

Q5: 5. What error occurs when attempting to DELETE a record from a Parent table that has matching child records in a Child table?

A) ORA-00937 Not a single-group function
B) ORA-02292 Integrity constraint violated - child record found
C) ORA-00904 Invalid identifier
D) ORA-01400 Cannot insert NULL

Explanation: Referential integrity blocks deletion of parent records while dependent child records exist.

Q6: 6. What is the result of the expression: NULL = NULL in SQL?

A) TRUE
B) FALSE
C) UNKNOWN
D) NULL

Explanation: Comparisons with NULL evaluate to UNKNOWN in 3-valued SQL logic.

Q7: 7. To check if a column 'MiddleName' contains no value, which SQL condition must be used?

A) WHERE MiddleName = 'NULL'
B) WHERE MiddleName IS NULL
C) WHERE MiddleName = NULL
D) WHERE MiddleName EQUALS NULL

Explanation: NULL cannot be tested with '='. You must use 'IS NULL'.

Q8: 8. In SQL evaluation precedence, which operator has the highest priority among logical operators?

A) NOT
B) AND
C) OR
D) All have equal priority

Explanation: Operator priority order is: NOT (highest) -> AND -> OR (lowest).

Q9: 9. Which SQL DDL command is used to define a mandatory Foreign Key?

A) FOREIGN KEY (DeptId) REFERENCES DEPT(DeptId) NULL
B) DeptId NUMBER NOT NULL REFERENCES DEPT(DeptId)
C) MANDATORY FOREIGN KEY (DeptId)
D) DeptId REQUIRED DEPT(DeptId)

Explanation: Adding NOT NULL to the foreign key column constraint enforces mandatory relationship participation.

Q10: 10. What is a Weak Entity in ERD modeling?

A) An entity with no primary key that cannot be saved.
B) An entity whose existence depends on another strong entity (e.g., Enrolment depending on Student and Subject).
C) An entity with only one attribute.
D) A table with no foreign keys.

Explanation: Weak entities depend on strong entities for identity or existence.